OGUZCAN KARADENIZ

AI and Computer Vision Engineer

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EXPERIENCE

AI and Computer Vision Engineer

AIFIT

06/2023 - Present | London, United Kingdom

- Developed AI-powered end-to-end virtual shoe try-on system using advanced CNNs and Transformers with PyTorch and TensorFlow. Applied computer vision techniques such as Point Cloud Generation, Registration, Segmentation, PDM, 3D Shape Reconstruction, and SSA.
- Achieved 15% higher accuracy and reduced overfitting by applying transfer learning, hyperparameter tuning, and model ensembling techniques.
- Doubled inference performance and reduced cloud computing costs by 35% through model optimization strategies and architectural improvements. Built scalable MLOps pipelines using AWS services such as SageMaker for real- time monitoring, training, and model versioning.
- Created automated data pipelines with custom web scraping algorithms.
- Applied preprocessing and feature engineering techniques to uncover key patterns and enhance overall machine learning performance.
- Designed and deployed LLM-based AI agents using GPT-4, OpenAI function-calling, FAISS, and RAG architecture. Built a recommendation agent and analysis agent to enhance personalization and system automation. Deployed on AWS Lambda via FastAPI, reducing manualoperations by 60%.

Full Stack Web Developer

Freelance

08/2024 - Present | London, United Kingdom

- End-to-end Shopify app developer with hands-on experience in building secure, scalable, and real-time applications using Node.js, React, and AWS services. Developed custom storefront components, app proxy endpoints, and backend APIs that integrate seamlessly with the Shopify ecosystem. Built
- responsive frontend interfaces with React, while managing backend logic and data flow using modular Express.js architecture. Leveraged AWS Lambda, API Gateway, and DynamoDB to implement serverless, event-driven features and reduce infrastructure overhead. Focused on delivering production-ready applications that enhance user experience, automate workflows, and support real-time communication through WebSocket and Server-Sent Events (SSE).

University Project Coordinator

Bilgi University

03/2020 - 08/2020 | Istanbul, Turkey

- As a software and electronic system developer, I managed and guided the development of a UV-C based sterilization system for COVID-19.
- I developed and designed a sterilization device that quickly disinfects users' belongings at the entrance of shopping malls to promote public health safety.

Software Engineer Intern

SoftTech

06/2019 - 08/2019 | Istanbul, Turkey

- I created an IoT project with an ultra-low power NodeMCU that provides almost 1 year of battery life and designed an occupancy monitoring system for doors that can be installed inside.
- Together with Prof. Dr. Baykal Sarıoğlu, I had the chance to showcase the project at the Softtech conference and show its design and innovative functionality.

SUMMARY

I am an AI and computer vision engineer with strong experience in machine learning, computer vision and data science. I have developed several projects on real-time image and video processing. I have also developed web scraping algorithms and then analyzed the data applying data science steps to create large-scale datasets. I have optimized, trained and deployed models based on Transformers and CNN-based architectures, and built LLM-based recommendation agents. I am an expert in using popular AI frameworks (Pytorch, Tensorflow) and developing and deploying MLops pipelines using cloud technologies (AWS). I am ready to demonstrate my competence in machine learning, computer vision and data science to create intelligent systems, extract actionable insights from data and develop seamless solutions that will increase team performance.

SKILLS

Python | PyTorch | Matlab | OpenCV | Keras | TensorFlow | AWS | C | C++ | DevOps | Git | Docker | NumPy | Matplotlib | Pandas | Scikit-learn | System Design | Selenium | Scipy | ROS | Gazebo | AutoCAD | Java | MySQL | HTML | CSS | Javascript | PHP | Typescript | React | Node.js

PUBLICATIONS

Acknowledge of Emotions for Improving Student-Robot Interaction

The Science Press

04/2023

https://www.techscience.com/iasc/v37n1/52640

Facial Expression Recognition in the Wild with Application in Robotics

IEEE

10/2021

https://ieeexplore.ieee.org/document/9558909

EMRES: A New EMotional RESpondent Robot

IEEE

10/2021

https://ieeexplore.ieee.org/document/9576513

Software Engineer Intern

Sunny Electronic Inc

06/2018 - 08/2018 Istanbul, Turkey

- Developed a server-client project for internal messaging between employees in the company.
- Enabled messaging over the company network via computer and Android phones in the R&D department.

EDUCATION

Computer Vision, Robotics and Machine Learning MSc

University of Surrey

Merit

02/2022 - 02/2023 Guildford, United Kingdom

- Dissertation: Object Detection and Tacking for Autonomous Driving with YOLOv7 with KITTI and Waymo datasets.

Electrical and Electronics Engineering BSc

Istanbul Bilgi University

GPA **3.13** / 4.0

09/2016 - 09/2020 Istanbul, Turkey

- Dissertation: Facial Expression Recognition Robot powered cloud GPU and powered nvidia jetson nano

CERTIFICATIONS (LINK)

Neural Networks and Deep Learning

Convolutional Neural Networks

Machine Learning with Python

CS50's Introduction to Computer Science

Introduction to Self-Driving Cars

State Estimation and Localization for Self-Driving Cars

LANGUAGES

English
Proficient

Turkish
Native

INTERESTS

Karting Volleyball Tenis